Distribution Models and Portfolio Risk Management: Evidence from Technology and Energy Stocks

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Abstract

Can protecting a portfolio against losses become more costly than the risk it seeks to avoid? I investigate this question using daily returns for 20 technology and energy equities from 2021–2025. Normal, location-scale Student-t, and two-component Gaussian mixture models are fitted on 2021–2023 data. Student-t achieves the lowest Bayesian information criterion (BIC) for all 20 stocks. Yet that clean statistical victory does not produce a consistent advantage in 2024–2025 out-of-sample value-at-risk (VaR) and expected shortfall (ES) comparisons.

The economically decisive result comes from a 502-day portfolio backtest. A fully invested benchmark earns 91.60%, compared with only 21.49–24.17% for three volatility-scaled, VaR-based position-sizing strategies. The risk controls reduce maximum drawdown from 20.72% to 6.09–6.78%, but also reduce average equity exposure to 33.15–36.98%. They fail to improve risk-adjusted returns: their annualized Sharpe ratios are approximately 1.40, below the benchmark's 1.55. The Student-t strategy has the lowest volatility and drawdown, but also the lowest exposure and cumulative return. A supplementary theoretical European call study illustrates that distributional assumptions can affect valuation without establishing a market-pricing advantage.

The investment lesson is sharper than a model-ranking exercise. In this particular rising-market test, the largest realized cost of conservative VaR sizing was the upside participation surrendered. Lower drawdown alone was not evidence of a better investment outcome. Risk controls should therefore be evaluated against their opportunity cost, not simply against the losses they prevent. These historical results do not establish that remaining fully invested maximizes future returns or the probability of profit.

Keywords: Portfolio risk management, Opportunity cost, Market exposure, Student-t, Value-at-risk, Expected shortfall

1. Introduction

Investment risk management promises to limit losses. Research on volatility timing and volatility-managed portfolios has shown that adjusting market exposure can have economic value, but that value depends on the strategy and evaluation setting [1, 2]. Yet every decision to reduce market exposure has a second effect: it also reduces participation when prices rise. The relevant question for an investor is therefore not only whether a model can estimate downside risk, but whether acting on that estimate improves the eventual balance between losses avoided and gains forgone. A smoother wealth curve is not automatically a more profitable investment strategy.

This study asks a direct question: can a sophisticated risk model protect a portfolio while leaving its investor worse off in total and risk-adjusted returns? I address it by tracing three links in the modeling chain. First, does a flexible distribution fit historical stock returns better than a Normal benchmark? Second, does that improvement survive out-of-sample tests of tail-risk forecasts? Third, when risk estimates determine the invested fraction, do they improve the financial result or primarily shrink market participation?

I examine 20 US-listed equities. The technology group comprises MSFT, AAPL, NVDA, GOOGL, META, AMZN, AMD, INTC, ORCL, and CRM. The energy group comprises XOM, CVX, COP, EOG, OXY, SLB, HAL, VLO, MPC, and PSX. These are descriptive business groupings rather than a strict industry classification. I fit Normal, Student-t, and two-component Gaussian mixture distributions using 2021–2023 data and reserve 2024–2025 for out-of-sample tests. The use of non-Gaussian return distributions is motivated by long-standing evidence of departures from normality in financial returns [3, 4]. A supplementary option-pricing exercise explores a separate consequence of distribution choice, but the central contribution is the investment decision implied by the risk models.

The results set up an instructive contrast. Student-t wins the BIC comparison for every stock, but no distribution wins consistently across portfolio VaR and ES criteria. More strikingly, the fully invested portfolio returns 91.60% during the test period, while the three VaR-sized strategies return 21.49–24.17%. Their drawdowns are much smaller, but their Sharpe ratios are also lower. This makes the opportunity cost of de-risking the central economic result, rather than the identity of the best-fitting distribution.

The claim is deliberately conditional on the observed market. The backtest does not estimate the probability that a fully invested strategy will win in a future market, and it does not isolate the causal effect of the distribution from the effect of lower average exposure. It does, however, demonstrate why a risk-control rule must justify both its downside protection and the returns it sacrifices.

2. Data and Research Design

2.1. Return construction and sample split

Adjusted daily closing prices are collected from January 2021 through December 2025. For stock i, the daily log return is

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right). \tag{1}$$

Returns are expressed in percentage points when fitting the daily distributions. For the risk analysis, portfolio daily simple returns are expressed as decimal fractions. Keeping these conventions separate avoids an unintended factor-of-100 error.

The training period covers 2021–2023. The 2024–2025 period is reserved for out-of-sample evaluation. Model parameters and fixed risk thresholds are obtained from training data rather than refitted using test-period observations. The test set contains 502 trading days.

For descriptive tail comparisons, standardized returns are

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_i}{s_i}, \tag{2}$$

where $\bar{r}_i$ and s_i are calculated within the period being described. These standardized observations summarize historical extremes rather than prospective probabilities.

2.2. Candidate return distributions

I fit three distributions to each stock's daily log returns by maximum likelihood. The Normal density is

$$f_N(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{3}$$

The location-scale Student-t density is

$$f_t(x) = \frac{1}{s} t_\nu\left(\frac{x-m}{s}\right), \tag{4}$$

where ν controls tail thickness. Smaller ν corresponds to heavier polynomial tails. The two-component Gaussian mixture is

$$f_{\text{GMM}}(x) = w\phi(x;\mu_1,\sigma_1^2) + (1-w)\phi(x;\mu_2,\sigma_2^2). \tag{5}$$

The Normal, Student-t, and GMM specifications have 2, 3, and 5 fitted parameters, respectively. The GMM is a static mixture rather than a time-varying regime model.

To balance fit against parameter count, I use the Bayesian information criterion (BIC) [5],

$$\text{BIC}_M = k_M \log N - 2\sum_{t=1}^{N} \log f_M(r_t). \tag{6}$$

Lower BIC is preferred. For comparisons across stocks, I define

$$\Delta\text{BIC}_{i,M} = \text{BIC}_{i,M} - \min_{M'} \text{BIC}_{i,M'}. \tag{7}$$

The same training observations are used to compare the three distributions for each stock.

3. Statistical Fit: A Clear Winner, but Not Yet an Investment Edge

3.1. Visual comparison and model selection

Figure 1 displays six examples of training-period return histograms and fitted densities. The Student-t curve often follows the sharp central peak more closely than the Normal curve. The models nevertheless differ by stock, and visual inspection alone cannot establish a ranking.

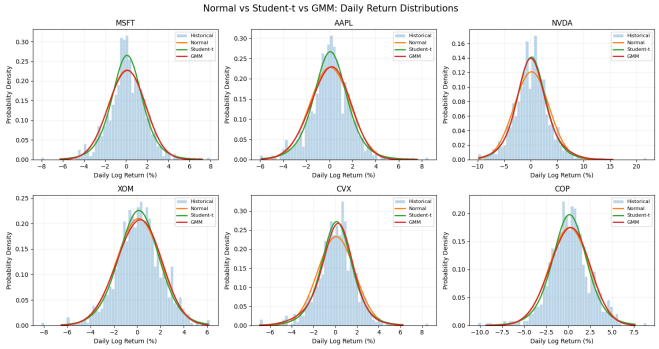


Figure 1: Training-period daily log-return histograms and fitted Normal, Student-t, and two-component GMM densities for six selected stocks, 2021–2023. Returns are expressed in percent. These examples illustrate the fitted shapes and are not a separate test sample.

Student-t achieves the lowest training-period BIC for all 20 stocks. The second-place ranking varies across the two groups. GMM ranks ahead of Normal for eight of the ten technology stocks, while Normal ranks ahead of GMM for seven of the ten energy stocks.

Group	t best	GMM 2nd	Normal 2nd
Technology (10)	10	8	2
Energy (10)	10	3	7
Total (20)	20	11	9

Table 1: Training-period BIC ranking counts across 20 stocks.

The strength of the Student-t advantage is not uniform. For XOM, HAL, and MPC, the BIC differences between Normal and Student-t are only 2.3, 1.7, and 1.7, respectively. For META and ORCL, GMM is substantially closer to Student-t than Normal is. This variation matters when moving from a statistical fit to a financial application. A common winner across stocks does not imply a common economic effect.

3.2. Tail characteristics

Figure 2 reports the fitted Student-t degrees of freedom for all 20 stocks. The technology group has a mean ν of 4.898 and median of 4.558. The corresponding energy values are 8.370 and 8.152. The lower fitted degrees of freedom for technology stocks indicate heavier tails within the chosen Student-t specification.

As a separate descriptive check, I count observations beyond three sample standard deviations. Under a standard Normal

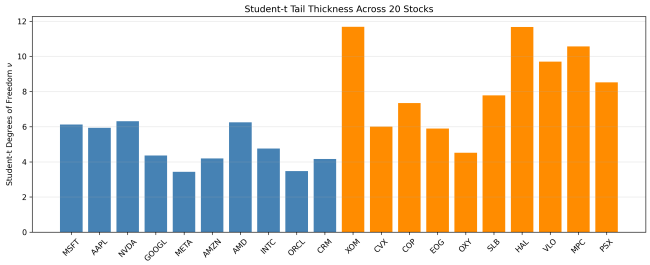


Figure 2: Maximum-likelihood Student-t degrees of freedom for the 20 stocks, fitted on 2021–2023 returns. Smaller ν indicates heavier fitted tails.

benchmark, the probability in either one-sided tail is approximately 0.135%. In the training sample, the technology group's mean frequencies are 0.585% on both sides. The energy group averages 0.532% in the left tail and 0.279% in the right tail. These values are calculated stock by stock and then averaged across each group.

The Student-t degrees of freedom summarize a symmetric fitted distribution. They do not capture every aspect of the observed left-right imbalance. The group comparisons are also descriptive. The stocks share market shocks, so they cannot be treated as 20 independent replications of a sector-level experiment.

4. Out-of-Sample Risk Forecasts: The Advantage Weakens

4.1. VaR and ES evaluation

The next question is whether a model that fits the training returns well also provides useful estimates of future portfolio losses. Tail-risk estimation is especially challenging when financial returns are heteroscedastic and heavy-tailed [6]. The portfolio risk exercise evaluates the same underlying portfolio over the 2024–2025 test period. Model-specific VaR and ES thresholds are estimated from the training period and held fixed during testing. This is a static forecast comparison, not an adaptive volatility model.

Let R_t denote a portfolio daily simple return and $L_t = -R_t$ its loss. For confidence level α, VaR is the α quantile of the loss distribution,

$$\mathrm{VaR}_\alpha = \inf\{l : \Pr(L \le l) \ge \alpha\}. \tag{8}$$

For a continuous loss distribution, expected shortfall is the mean loss conditional on exceeding that quantile,

$$\mathrm{ES}_\alpha = E[L \mid L \ge \mathrm{VaR}_\alpha]. \tag{9}$$

The general quantile-integral definition is used when the distribution is not continuous. Expected shortfall is closely related to the coherent risk-measure framework and conditional value-at-risk optimization [7, 8].

A VaR exceedance occurs when the realized loss exceeds the model's VaR threshold. Under correct unconditional coverage, the expected number of exceedances across N days is $N(1-\alpha)$. Here $N = 502$, giving 25.10 expected exceedances at 95% confidence and 5.02 at 99%.

4.2. Exceedance counts and realized losses

Figure 3 compares observed exceedance counts with their nominal expectations. At 95% confidence, Normal and GMM each record 12 exceedances and Student-t records 14. All three counts are below the expected 25.10. The fixed thresholds therefore appear conservative over this particular test period.

At 99% confidence, Normal records six exceedances, Student-t four, and GMM five. These are all close to the expected 5.02. With so few extreme observations, a difference of one or two exceedances is not strong evidence that one model is statistically superior. Formal backtesting methods distinguish unconditional coverage from the independence of exceedances [9, 10].

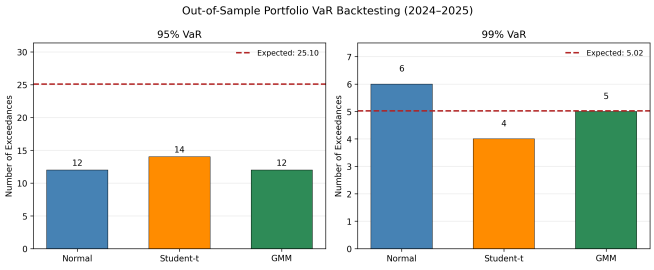


Figure 3: Out-of-sample VaR exceedance counts for 502 portfolio trading days in 2024–2025. Dashed horizontal lines indicate expected counts under nominal 95% and 99% coverage.

Figure 4 places the three fixed 99% VaR thresholds alongside realized daily portfolio returns. The predicted loss thresholds are approximately 3.86% for Normal, 4.30% for Student-t, and 3.96% for GMM. The largest negative observations are concentrated in a small number of trading days, including a pronounced episode in spring 2025. The plot suggests that the volatility environment changes over time, although clustering should be evaluated formally rather than inferred from a picture alone. Conditional heteroscedasticity provides a well-established framework for studying such variation [11].

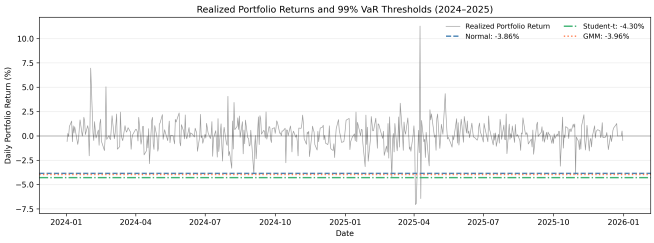


Figure 4: Realized daily simple portfolio returns in 2024–2025 and fixed, training-based 99% VaR return thresholds. A return below a model's horizontal threshold is a VaR exceedance.

4.3. Predicted versus realized VaR and ES

Figure 5 compares model-based loss estimates with empirical summaries of the test-period loss distribution. The realized 95% VaR is approximately 1.95%, compared with model estimates of 2.58–2.74%. The realized 95% ES is 3.15%, compared with model estimates of 3.40–3.68%.

At 99% confidence, the realized VaR is approximately 3.91%. GMM is closest at 3.96%, followed by Normal at 3.86% and Student-t at 4.30%. The realized 99% ES is approximately 5.45%. Student-t is closest at 5.60%, while GMM and Normal give lower estimates of 4.57% and 4.44%.

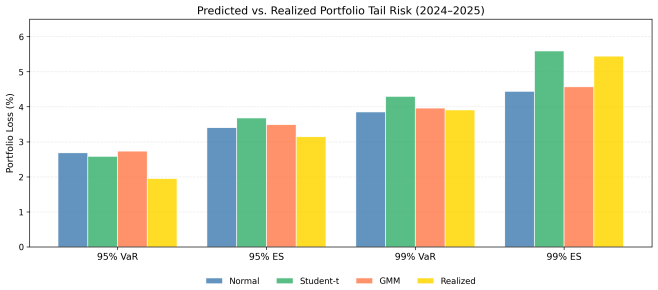


Figure 5: Model-predicted portfolio VaR and ES compared with empirical test-period loss quantiles and tail means, 2024–2025. Values are percentages of portfolio value. Realized values are retrospective summaries rather than forecasts.

	95% VaR	95% ES	99% VaR	99% ES
Normal	2.685	3.403	3.856	4.439
Student-t	2.581	3.683	4.298	5.595
GMM	2.736	3.489	3.965	4.572
Realized	1.950	3.147	3.913	5.448

Table 2: Model-based portfolio tail-risk estimates and empirical test-period benchmarks, in percent.

These results do not identify one universally best model. Student-t provides the closest 95% VaR and 99% ES estimates, while GMM is closest to the realized 99% VaR and Normal is closest to the realized 95% ES. The empirical tail estimates are themselves noisy, especially at 99% confidence, where only a handful of days determine the outcome. A formal comparison would require coverage and exceedance-independence tests, alongside an evaluation of the statistical uncertainty in ES estimates.

5. The Investment Test: The Opportunity Cost of Protection

5.1. Position-sizing rule and implementation

The preceding results compare the statistical quality of fixed risk thresholds. A separate backtest asks what happens when those estimates guide an investment decision. The economic question is related to earlier studies of volatility timing and volatility-managed portfolios [1, 2]. I use the same underlying portfolio in all four strategies and compare a fully invested benchmark with Normal-, Student-t-, and GMM-based position sizing. The backtest covers the 502 trading days in 2024–2025.

The fitted 1% portfolio return quantile q_M for each model M comes from the 2021–2023 training sample. During the test period, I scale the distance between that quantile and the fitted model center m_M by a volatility multiplier,

$$v_t = \mathrm{SD}(R_{t-63}, \ldots, R_{t-1}), \tag{10}$$

$$h_t = \mathrm{clip}\left(\frac{v_t}{s_{\mathrm{train}}}, 0.25, 4\right), \tag{11}$$

$$q_{M,t} = m_M + h_t(q_M - m_M), \tag{12}$$

$$\widehat{\mathrm{VaR}}^{99\%}_{M,t} = \max(-q_{M,t}, 10^{-8}), \tag{13}$$

$$a_{M,t} = \min\left(1, \frac{0.01}{\widehat{\mathrm{VaR}}^{99\%}_{M,t}}\right). \tag{14}$$

Here s_{train} is the training-period standard deviation of the portfolio's simple returns, and $a_{M,t}$ is the fraction of capital invested in the underlying portfolio. The rolling standard deviation uses only returns through day $t-1$, so the exposure for day t does not use that day's realized return. The maximum exposure is 100%, with no leverage. The remaining capital earns zero in this experiment. The benchmark holds $a_t = 1$ throughout.

Transaction costs are estimated at five basis points per unit of absolute exposure change. The daily net strategy return and accumulated wealth are

$$R^{\mathrm{net}}_{M,t} = a_{M,t}R_t - 0.0005\,|a_{M,t} - a_{M,t-1}|, \tag{15}$$

$$W_{M,t} = \prod_{u \le t}(1 + R^{\mathrm{net}}_{M,u}). \tag{16}$$

The initial move from cash to the chosen exposure is charged as turnover. This is a dynamic volatility-scaled strategy, not a rolling re-estimation of the three return distributions. It also differs from the static VaR backtest in the preceding section.

5.2. Wealth trajectories, foregone gains, and performance

Figure 6 shows the resulting net wealth paths. The fully invested portfolio grows from 1 to 1.916, corresponding to a cumulative return of 91.60%. Its path is considerably more volatile, including a sharp drawdown around spring 2025. All three VaR-controlled portfolios follow much smoother paths and end between 1.215 and 1.242. Their limited participation in the underlying portfolio's rally is the visible cost of reducing exposure.

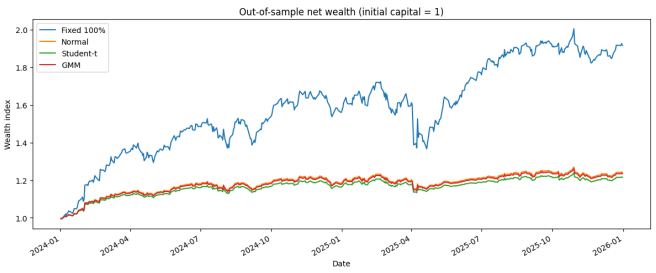


Figure 6: Out-of-sample net wealth of the fully invested portfolio and three volatility-scaled, VaR-based position-sizing strategies, 2024–2025. Initial wealth is one. Returns include estimated exposure-change transaction costs. Cash earns zero.

Strategy	Return (%)	Vol. (%)	Max DD (%)	Sharpe
Fixed 100%	91.60	22.77	-20.72	1.548
Normal	24.17	8.00	-6.78	1.399
Student-*t*	21.49	7.17	-6.09	1.399
GMM	23.45	7.78	-6.60	1.399

Strategy	99% VaR (%)	99% ES (%)	Avg. exposure (%)
Fixed 100%	3.91	5.45	100.00
Normal	1.38	1.80	36.98
Student-*t*	1.24	1.62	33.15
GMM	1.35	1.75	35.96

Table 3: Out-of-sample strategy performance, 2024–2025. VaR and ES are empirical 99% daily loss measures of strategy net returns. Annualized volatility and Sharpe ratios assume zero cash return.

Strategy	Total turnover	Estimated cost
Fixed 100%	1.0000	0.050%
Normal	3.1061	0.155%
Student-*t*	2.7765	0.139%
GMM	3.0158	0.151%

Table 4: Exposure turnover and estimated transaction costs. Turnover is the sum of absolute daily changes in exposure, including the initial investment. Estimated costs are fractions of initial capital under the backtest convention.

5.3. What the investor actually gave up

The headline comparison is not a small statistical difference. Relative to the 91.60% fully invested return, the Normal, Student-*t*, and GMM strategies surrender 67.43, 70.11, and 68.15 percentage points of cumulative return, respectively. Expressed as a fraction of the benchmark's cumulative gain, the forgone gain is approximately 74%, 77%, and 74%. These are descriptive differences in realized backtest outcomes, not estimates of the gains that would be sacrificed in an average future market.

The other side of the trade-off is real. The Student-*t* rule cuts the magnitude of maximum drawdown by 14.63 percentage points, from 20.72% to 6.09%, and reduces annualized volatility by 15.60 percentage points. An investor facing strict loss limits might rationally prefer that smoother path. But an investor whose objective is to maximize growth or the observed Sharpe ratio would not have benefited from these particular risk controls over this sample.

The exposure figures help explain the contrast. The VaR strategies invest only about one-third of capital in equities on average, with the remainder earning zero in the assumed cash account. This mechanical reduction in participation is an important alternative explanation for their lower volatility and smaller drawdowns. The experiment cannot establish how much of the protection comes from successful timing rather than from holding less equity. Nevertheless, it makes the economic trade-off visible: risk was reduced by accepting substantially less market participation.

The risk-controlled strategies lower annualized volatility from 22.77% to 7.17–8.00% and maximum drawdown from 20.72% to 6.09–6.78%. Student-*t* produces the smallest realized losses, but it also has the smallest average market exposure at 33.15%. Its lower risk cannot be attributed to superior risk timing without controlling for this difference in invested capital. The Normal and GMM strategies take slightly more risk and earn slightly higher total returns.

The three model-based strategies have almost indistinguishable annualized Sharpe ratios, around 1.399, below the fully invested benchmark's 1.548. Their near-identical risk-adjusted results are consistent with the fact that they invest in the same underlying portfolio and vary mainly in exposure. The evidence therefore supports a narrower conclusion than "Student-*t* improves investment performance." VaR-based position sizing reduces risk in this sample, but the improvement is accompanied by lower participation in gains, and the historical BIC winner does not deliver a clear advantage in risk-adjusted performance. A matched-exposure or matched-risk-budget experiment would be needed to isolate the incremental value of the distribution model.

5.4. Did model sophistication pay for itself?

The distinction between the three VaR rules is much smaller than the distinction between any VaR rule and full investment. Their annualized Sharpe ratios differ by only about 0.0005 using unrounded values, despite their very different distributional assumptions and Student-*t*'s universal BIC lead. The more elaborate historical fit therefore does not translate into an observable Sharpe improvement within this experiment.

This is not a general argument against downside protection. It is a warning against treating lower realized volatility or a better in-sample likelihood as sufficient evidence of a superior investment rule. The tested risk budget, the no-leverage constraint, the volatility-scaling formula, and the positive returns of the underlying portfolio during 2024–2025 all contribute to the observed opportunity cost. A different market regime or investment objective could reverse the preference. The conclusion supported here is specific and actionable: when judging a risk overlay, report both the protection obtained and the return and market exposure surrendered.

6. Additional Evidence: Distribution Assumptions and Option Values

6.1. Why an unmodified Student-t model is problematic

The strong historical fit of Student-*t* raises a separate question about its use for option valuation. For a one-period log return X,

$$S_{t+\Delta t} = S_t e^X. \tag{17}$$

A risk-neutral stock-price model requires a finite expected growth factor. For Gaussian $X \sim N(\mu, \sigma^2)$,

$$E[e^X] = \exp\left(\mu + \frac{\sigma^2}{2}\right) < \infty. \tag{18}$$

A Student-*t* density instead has a polynomial positive tail, $f_t(x) \sim Cx^{-(\nu+1)}$. The factor e^x dominates this decay, and

$$E[e^X] = \int_{-\infty}^{\infty} e^x f_t(x)\, dx = \infty. \tag{19}$$

Adding a finite drift shift does not fix this divergence. Figure 7 illustrates the different behavior of the exponential-moment integrands.

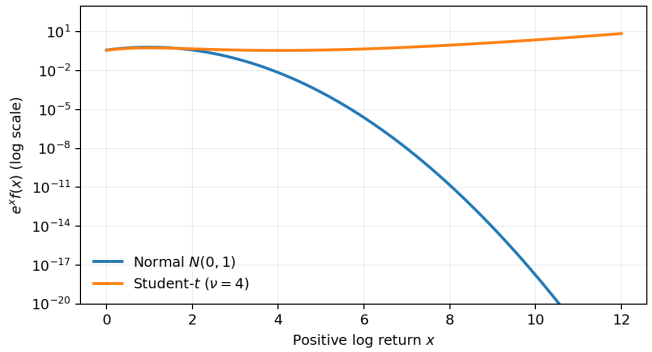


Figure 7: Positive-tail exponential-moment integrand $e^x f(x)$ for standard Normal and Student-t ($\nu = 4$) densities. The logarithmic vertical axis highlights the divergence of the untruncated Student-t exponential moment.

To construct a finite-moment illustration, I truncate the fitted Student-t density to finite bounds L and U,

$$f_{\text{tr}}(x) = \frac{f_t(x)}{F_t(U) - F_t(L)}, \qquad L \le x \le U. \tag{20}$$

The finite exponential moment under the truncated density is

$$E[e^{X^{\text{tr}}}] = \frac{\int_L^U e^x f_t(x)\,dx}{F_t(U) - F_t(L)} < \infty. \tag{21}$$

The risk-neutral log-return increment is then specified as

$$c_{\text{tr}} = \log E[e^{X^{\text{tr}}}], \tag{22}$$

$$R^Q = (r_f - q)\Delta t + X^{\text{tr}} - c_{\text{tr}}. \tag{23}$$

This construction satisfies $E^Q[e^{R^Q}] = e^{(r_f - q)\Delta t}$. It is one modeling choice, not a unique risk-neutral measure implied by historical data. Truncation can materially change the right tail that matters for call payoffs.

6.2. *Option price comparisons*

I compare theoretical European call prices for META and XOM at maturities of 21 and 126 trading days and strike-to-spot ratios of 0.9, 1.0, and 1.1. The benchmark is Black–Scholes [12], alongside Gaussian Monte Carlo, GMM, and truncated Student-t simulations. All model-based call prices are interpreted through

$$C_0 = e^{-r_f T} E^Q[(S_T - K)^+]. \tag{24}$$

Figure 8 shows the normalized call prices.

Gaussian Monte Carlo closely tracks Black–Scholes, as expected when their Gaussian dynamics and parameters are consistent. GMM prices are also generally close to the benchmark.

The truncated Student-t prices are noticeably lower for META, but remain much closer to Black–Scholes for XOM. Figure 9 quantifies the difference as

$$\Delta C = 100 \frac{C_{\text{model}} - C_{\text{BS}}}{C_{\text{BS}}}. \tag{25}$$

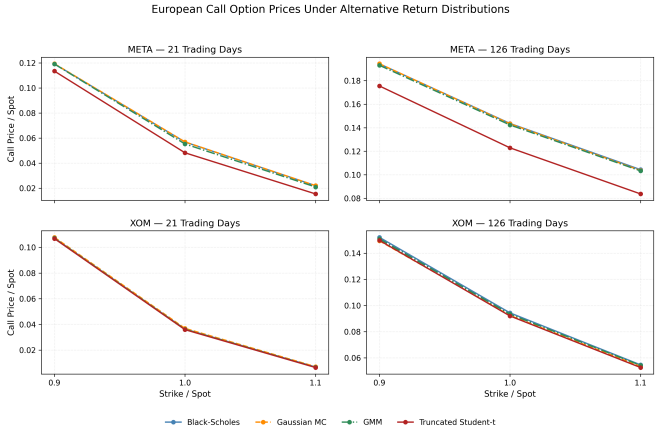


Figure 8: Normalized theoretical European call prices for META and XOM under Black–Scholes, Gaussian Monte Carlo, GMM, and truncated Student-t assumptions. Maturities are 21 and 126 trading days.

For META's 21-day out-of-the-money call, the truncated Student-t price is 29.91% below Black–Scholes. The corresponding difference for XOM is 5.87%. At 126 days, the out-of-the-money differences are 19.75% and 3.75%, respectively.

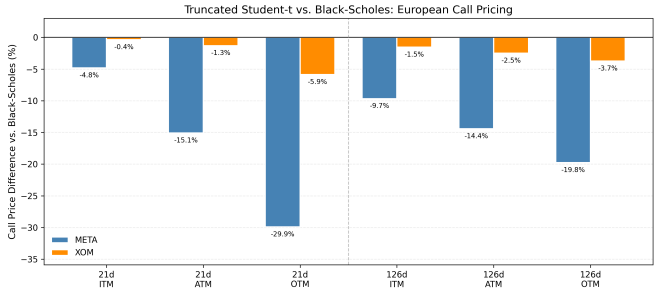


Figure 9: Relative differences between truncated Student-t and Black–Scholes theoretical call prices. Negative values indicate lower prices under the truncated Student-t construction. ITM, ATM, and OTM refer to strike-to-spot ratios of 0.9, 1.0, and 1.1.

These results should not be read as evidence that heavier tails always lower call prices. The fitted center and scale, the finite truncation bounds, and the drift adjustment all affect the simulated right-tail payoffs. Nor do differences from Black–Scholes establish pricing errors against market option quotes. The exercise demonstrates sensitivity to assumptions, not market pricing accuracy.

7. Discussion and Limitations

The study's most important distinction is between protecting capital and improving an investment outcome. Student-t is unambiguously preferred by training-period BIC, but its out-of-sample tail-risk performance is mixed. More importantly, its lower realized strategy risk is accompanied by the lowest average invested fraction. A model that induces a smaller position can generate a smaller drawdown without providing a better prediction of when losses will occur. The nearly identical

Sharpe ratios across the three risk overlays reinforce the absence of a clear risk-adjusted advantage in this particular test.

The market setting is essential to the interpretation. Unlike studies finding economic benefits from volatility-based timing rules [1, 2], this particular backtest finds no improvement in the observed Sharpe ratio. These differences should not be interpreted as a direct replication test because the assets, timing rules, and constraints differ. The fully invested portfolio experienced a large cumulative gain over 2024–2025. In such a sample, spending long periods at roughly one-third equity exposure creates a substantial opportunity cost. The result should not be extrapolated into a claim that full investment is optimal in downturns or that it has a higher future probability of profit. No future profit probability is estimated here, and the reported Sharpe ratios are descriptive rather than statistically significant differences.

The analysis covers 20 selected stocks and one train-test split. The static VaR coverage study uses fixed training thresholds, whereas the separate portfolio strategy applies a lagged 63-day volatility multiplier to training-period distribution quantiles. The strategy outcomes depend on the 1% VaR budget, exposure cap, transaction-cost assumption, and zero return on uninvested cash. The models' different average exposures prevent a causal attribution of strategy performance to distribution quality alone. At 99% confidence, the realized ES estimate also rests on very few extreme observations. The supplementary option prices are theoretical outputs from assumed risk-neutral constructions, not tests against market option quotes. These boundaries define the claims rather than invalidating the observed comparisons.

8. Conclusion

The most expensive risk control in this backtest was the market exposure it gave up. The fully invested portfolio earned 91.60% during 2024–2025. Three VaR-driven overlays reduced maximum drawdown from 20.72% to roughly 6–7%, but returned only 21–24%. The price of smoother wealth was approximately 67–70 percentage points of cumulative return relative to the benchmark. None of the overlays improved the observed Sharpe ratio.

The model comparison sharpens this result. Student-t won all 20 historical BIC comparisons and generated the lowest realized volatility and drawdown in the portfolio test. Yet it also maintained the lowest equity exposure and earned the lowest cumulative return among the risk-controlled strategies. The three overlays produced virtually identical Sharpe ratios. Better historical distribution fitting therefore did not deliver a detectable risk-adjusted investment edge in this setting.

The investment implication is not to abandon risk management or always remain fully invested. It is to reject the assumption that minimizing risk automatically maximizes financial value. A risk overlay earns its place only when the protection it provides is worth the upside participation it sacrifices. For an investor prioritizing growth in the rising market studied here, the tested conservative VaR rules were an expensive form of protection. For an investor prioritizing drawdown limits, that same protection may have been worth paying for. The decision depends on the investor's objective, not on the sophistication of the fitted return distribution.

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